Zonoth

SESSION 05 · 2H

Kubernetes: architecture, kubectl, Pods

From one machine running Compose to a cluster that keeps Pinboard running on its own — and the one command you drive it with.

Why orchestration | Cluster architecture | API & kubectl | Pods in depth | Lab 05

SESSION 05

What we'll cover today

1 · Cluster architecture

Why orchestration, what Kubernetes is and isn't, and every process on the control plane and the nodes.

2 · The API & kubectl

The object model, desired state and reconciliation, kubectl grammar, kubeconfig, contexts and namespaces.

3 · Pods in depth

Shared network and volumes, lifecycle and status, requests/limits and QoS, init and sidecar containers.

4 · Lab 05

Create the kind cluster, load `pinboard-api:1.0` and run it as a Pod in the `pinboard` namespace.

01

Why Kubernetes

Compose gave us one host. Production needs many — and something that watches them for us.

MOTIVATION

What Compose could not do

Many hosts

Compose schedules onto **this** machine. Nothing spreads Pinboard over a fleet.

Self-healing

If the host dies, the containers die with it. Nobody restarts them elsewhere.

Rolling updates

`up -d` replaces containers all at once — a visible gap in service.

Scaling

`--scale api=3` is manual and capped by one machine's CPU.

Secrets & config

A `.env` file on a laptop is not a secret management story.

Traffic routing

No load balancing, no service discovery beyond a single bridge network.

DEFINITION

What Kubernetes is — and isn't

It is

- A **container orchestrator**: it places containers on machines and keeps them there
- A **declarative API** you POST desired state to
- A set of **controllers** that reconcile reality toward that state
- Extensible — CRDs, operators, CNI/CSI/CRI plugins

It is not

- A PaaS — it does not build your code or run a CI pipeline
- A container runtime — that is containerd, via the CRI
- A database, a message bus or a logging stack
- Free of operations — it moves the work, it does not delete it

Kubernetes 1.37 (kind v0.33) is the baseline for this course. The project ships three minor releases a year and supports the last three.

WHERE IT RUNS

Local clusters vs managed clusters

LOCAL — FOR LEARNING & CI	MANAGED — FOR PRODUCTION
kind — nodes are Docker containers; multi-node, fast, disposable. Our choice.	**AKS · EKS · GKE** — the cloud runs and upgrades the control plane for you.
minikube — VM or Docker driver, rich addons	You still own nodes, upgrades, quotas and cost
k3d (k3s in Docker) · Docker Desktop single-node	Real LoadBalancers, real StorageClasses, cloud IAM

✓ The API is identical everywhere. What changes is what a `LoadBalancer` Service or a `PersistentVolumeClaim` resolves to.

02

Cluster architecture

A control plane that decides, and nodes that do. Nine processes, and you only ever talk to one of them.

THE BRAIN

Control plane components

kube-apiserver

The only door into the cluster. REST over HTTPS, authn/authz, admission, validation.

etcd

Consistent key-value store — the single source of truth. Back it up to rebuild.

kube-scheduler

Watches for Pods with no node, scores candidates, writes the binding back to the API.

kube-controller-manager

Dozens of control loops in one binary — Deployment, Node, Job…

cloud-controller-manager

Cloud-specific loops: LoadBalancers, routes, node lifecycle. Not on kind.

CoreDNS (add-on)

Cluster DNS. Not strictly control plane, but every cluster has it — Session 07.

THE MUSCLE

Node components

kubelet

The node agent. Watches the API for Pods bound to its node, asks the runtime to start them, runs probes, reports status back.

kube-proxy

Programs iptables/IPVS (or is replaced by an eBPF CNI) so Service IPs reach Pod IPs.

Container runtime

containerd via the **CRI**, using runc and OCI images — the same images you built in Session 02.

⚠ **Dockershim was removed in Kubernetes 1.24.** "Kubernetes dropped Docker" only ever meant the shim — your Docker-built OCI images run unchanged on containerd.

FOLLOW THE REQUEST

What happens when you apply a Pod

01 — kubectl apply
YAML → JSON, POST to the apiserver

→

02 — apiserver
Authn, authz, admission, validate

→

03 — etcd
Object persisted, `nodeName` empty

→

04 — scheduler
Picks a node, writes the binding

→

05 — kubelet
Pulls image, CRI starts it, reports Running

Every arrow is a **watch** on the API, not a direct call. No component ever phones another — they all react to the same store.

LIVE DEMO · 8 MIN

Build a cluster and look inside it

Create the three-node course cluster with kind, then prove that the control plane is itself just Pods running on Kubernetes.

What you'll see · nodes, control-plane Pods, containerd as the runtime **Files** · labs/kind-config.yaml **Duration** · 8 min

03

The API and kubectl

Every object has the same four fields. Every command has the same three words.

Four fields, every single time

- **apiVersion** — group and version, e.g. `v1`, `apps/v1`
- **kind** — the type: Pod, Deployment, Service
- **metadata** — name, namespace, labels, annotations
- **spec** — **what you want**; you write this
- **status** — **what is**; the cluster writes this

YAML

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: pinboard-api
  namespace: pinboard
  labels:
    app.kubernetes.io/name: pinboard-api
spec:
  containers:
    - name: api
      image: pinboard-api:1.0
      imagePullPolicy: IfNotPresent
      ports:
        - { name: http, containerPort: 8080 }
```

THE CORE IDEA

Desired state and the reconciliation loop

01 — Declare
You write `spec` and apply it

→

02 — Observe
Controller watches actual state

→

03 — Diff
Compare desired vs actual

→

04 — Act
Create, delete or update to close the gap

↻

05 — Repeat
Forever — this is self-healing

Same idea as `terraform apply` — except nobody has to run it. The loop never stops, so drift is corrected in seconds.

ONE COMMAND TO LEARN

kubectl grammar

BASH

```
kubectl [verb] [resource] [name] [flags]

kubectl get pods                          # list
kubectl get pod pinboard-api -o yaml      # one object, full spec+status
kubectl describe pod pinboard-api         # human summary + events
kubectl logs pinboard-api -c api -f       # container logs, follow
kubectl exec -it pinboard-api -- sh       # a shell inside the container
kubectl port-forward pod/pinboard-api 8080:8080
kubectl delete pod pinboard-api
kubectl get pods -o wide --show-labels    # node, IP, labels
kubectl get pods -l app.kubernetes.io/name=pinboard-api
```

Verbs: `get describe create apply delete edit logs exec scale rollout`. Resources take short names (`po`, `deploy`, `svc`, `ns`) and are always plural-friendly.

TWO STYLES

Imperative vs declarative

Imperative — fast, unrepeatable

BASH

```
kubectl run pinboard-api \
  --image=pinboard-api:1.0 \
  -n pinboard --port=8080
```

Great for a five-second experiment. Nothing is written down; nobody can review it.

Declarative — the real workflow

BASH

```
kubectl apply -f pod.yaml
kubectl diff -f pod.yaml
kubectl apply -f k8s/ --dry-run=server
```

The file is the source of truth: reviewable, versioned, re-appliable, and the input to GitOps.

Best of both: `kubectl create deployment … --dry-run=client -o yaml > deploy.yaml` — generate the skeleton, then edit and commit it.

kubeconfig and contexts

- `~/.kube/config` — clusters, users, and **contexts** that pair them
- A context also carries a **default namespace**
- `$KUBECONFIG` can merge several files with `:`
- `kind create cluster` adds and selects `kind-pinboard`

> ⚠ Every production incident story starts with "I thought I was on dev." Put the context in your shell prompt — kube-ps1, starship, k9s.

BASH

```
kubectl config get-contexts
kubectl config current-context
# kind-pinboard

kubectl config use-context kind-pinboard

# pin the namespace for this context
kubectl config set-context --current \
  --namespace=pinboard

kubectl config view --minify
```

SCOPING

Namespaces

- A **name scope**, not a security boundary by itself
- Names must be unique **per namespace**; DNS becomes `svc.ns.svc.cluster.local`
- The hook for ResourceQuota, LimitRange, RBAC and NetworkPolicy
- Cluster-scoped objects have none: Node, PersistentVolume, StorageClass, Namespace

BASH

```
kubectl create namespace pinboard
kubectl get ns
# default   kube-system   kube-public
# kube-node-lease   pinboard

kubectl get pods -n pinboard
kubectl get pods -A          # all namespaces

kubectl api-resources --namespaced=false
```

✓ Ours is `pinboard` for every lab from here on.

SELF-SERVICE DOCS

kubectl explain — the API reference offline

BASH

```
kubectl explain pod.spec
kubectl explain pod.spec.containers.resources
kubectl explain deployment.spec.strategy --recursive

kubectl api-resources | head
# NAME          SHORTNAMES   APIVERSION   NAMESPACED   KIND
# pods          po           v1           true         Pod
# deployments   deploy       apps/v1      true         Deployment

kubectl api-versions | grep apps
```

`explain` reads the OpenAPI schema from **your** cluster, so it is always right for the version you are on — including CRDs your team installed.

LIVE DEMO · 10 MIN

A Pod from YAML, watched end to end

Side-load `pinboard-api:1.0` into kind, apply the manifest, and watch the events narrate scheduling, pulling and starting in real time.

What you'll see · explain · apply · events --watch · port-forward

Files · demos/session05/namespace.yaml · pod.yaml

Duration · 10 min

04

Pods in depth

The smallest thing Kubernetes will schedule — and the only thing that ever runs your containers.

ANATOMY

What a Pod actually shares

- **One network namespace** — one IP, one port space; containers reach each other on `localhost`
- **Volumes** — declared on the Pod, mounted per container
- **Lifecycle & node** — scheduled together, killed together
- **Not** shared: filesystem root, PID namespace (opt-in), memory limits

⚠ Two containers cannot both bind `:8080` in one Pod — same port space.



One app per Pod

`pinboard-api` and `pinboard-web` are **separate** Pods: they scale differently, fail differently and are released differently.

Extra containers belong in a Pod only when they are useless alone — a log shipper, a proxy, a config reloader.

READING THE ROOM

Pod lifecycle and the status you will see

STATUS	WHAT IT MEANS	FIRST MOVE
Pending	Accepted, not yet bound — no node fits, or a quota/taint blocks it	`describe pod` → Events
ContainerCreating	Node chosen; pulling the image, mounting volumes, wiring the network	Wait, then `describe`
Running	At least one container is up. **Ready** is a different column	`logs`, check READY `1/1`
CrashLoopBackOff	The container starts and exits repeatedly; kubelet backs off up to 5 min	`logs --previous`
ImagePullBackOff	Image missing, tag wrong, or registry auth failed	Check tag; `kind load`
Completed / Error	Process exited 0 or non-zero and `restartPolicy` allowed it to stop	Expected for Jobs

Requests, limits and QoS

- **requests** — what the **scheduler** reserves
- **limits** — what the **kubelet** enforces
- Over CPU → throttled. Over memory → **OOMKilled**

QOS CLASS	WHEN
Guaranteed	requests == limits, every container
Burstable	requests set, limits differ or absent
BestEffort	neither set — evicted first

YAML

```yaml
resources:
  requests:
    cpu: 50m          # 0.05 of a core
    memory: 32Mi
  limits:
    memory: 64Mi      # no cpu limit on purpose
```

Common production stance: **always** set memory request = limit, and leave CPU unlimited so bursts are absorbed instead of throttled.

Init containers and native sidecars

- **initContainers** run to completion, in order, before app containers start
- A **native sidecar** is an init container with `restartPolicy: Always` — on by default since 1.29, GA in 1.33
- Sidecars start **before** the app, keep running, and stop **after** it
- That ordering is what the old "sidecar as a normal container" hack never got right

YAML

```yaml
spec:
  initContainers:
    - name: wait-for-db
      image: busybox:1.37
      command: ["sh","-c",
        "until nc -z pinboard-db 5432; do sleep 1; done"]
    - name: log-shipper
      image: busybox:1.37
      restartPolicy: Always     # ← native sidecar
      command: ["sh","-c","tail -f /var/log/app.log"]
  containers:
    - name: api
      image: pinboard-api:1.0
```

DAY-TO-DAY

restartPolicy and working with a live Pod

- `restartPolicy: Always` (default) · `OnFailure` · `Never`
- It restarts **containers in place** — the Pod is never rescheduled
- A bare Pod on a dead node is simply gone. Nothing recreates it
- That is why real workloads are owned by a controller — Session 06

> ⚠ Never run a bare Pod in production. It is a debugging tool.

BASH

```
kubectl describe pod pinboard-api -n pinboard
kubectl logs pinboard-api -n pinboard -f
kubectl logs pinboard-api -n pinboard --previous
kubectl exec -it pinboard-api -n pinboard -- sh
kubectl port-forward pod/pinboard-api 8080:8080 \
  -n pinboard
curl -s localhost:8080/api/info
```

LAB 05 · 30–40 MIN

Pinboard meets Kubernetes

Create the three-node kind cluster, side-load the image you built in Lab 02, and run `pinboard-api:1.0` as a Pod in the `pinboard` namespace — first imperatively, then from a manifest you keep.

You'll practise: `kind create cluster` · `kind load docker-image` · `kubectl run` vs `apply` · namespaces & contexts · `port-forward` · `logs` · `describe` · requests/limits · reading an ImagePullBackOff · `kubectl explain`

Key takeaways

✓ One door, many loops

Everything goes through kube-apiserver into etcd; scheduler, controllers and kubelet just watch and react.

✓ spec is yours, status is theirs

Declare desired state and the reconciliation loop closes the gap — continuously, without you re-running anything.

✓ The Pod is the unit

Shared IP and volumes, requests/limits for scheduling, init and native sidecar containers for ordering.

✓ Next up

Session 06 — Deployments, rolling updates and the probes that make them safe.

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Questions?

Up next — Session 06: Workloads, rollouts and probes. Leave the kind cluster running.